Implicit Integration for Density of States

Luiz M. Faria[*], Ewen Lallinec[†], Antoine Levitt[†]

1 Periodic Hamiltonians and the density of states

This section introduces the quantum framework used throughout the paper.

1.1 Direct and reciprocal lattices

Let $d \in \{1, 2, 3\}$ be the spatial dimension. A *lattice* of $\mathbb{R}^d$ is a discrete subgroup of the form

$$\mathcal{R} = \left\{ A\mathbf{n} : \mathbf{n} \in \mathbb{Z}^d \right\} \cong \mathbb{Z}^d, \tag{1.1}$$

where $A = (\mathbf{a}_1 \cdots \mathbf{a}_d) \in \mathbb{R}^{d\times d}$ is non-singular. Its columns $\mathbf{a}_i$ are the *primitive vectors* of $\mathcal{R}$. The associated *unit cell* (or Wigner-Seitz cell) is the set of points of $\mathbb{R}^d$ that are closer to the origin than to any other lattice site:

$$\Gamma = \left\{ \mathbf{r} \in \mathbb{R}^d : \|\mathbf{r}\|_2 \leq \|\mathbf{r} - \mathbf{R}\|_2 \text{ for all } \mathbf{R} \in \mathcal{R} \setminus \{\mathbf{0}\} \right\}. \tag{1.2}$$

By construction, Γ tiles $\mathbb{R}^d$ under translation by $\mathcal{R}$.

The *reciprocal lattice* $\mathcal{R}^*$ is the set of vectors $\mathbf{G} \in \mathbb{R}^d$ such that $\mathrm{e}^{i\mathbf{G}\cdot\mathbf{R}} = 1$ for every $\mathbf{R} \in \mathcal{R}$. Equivalently,

$$\mathcal{R}^* = \left\{ B\mathbf{n} : \mathbf{n} \in \mathbb{Z}^d \right\}, A^\intercal B = 2\pi I. \tag{1.3}$$

The Wigner-Seitz cell of $\mathcal{R}^*$, called the *Brillouin zone*, is

$$\mathcal{B} = \left\{ \mathbf{k} \in \mathbb{R}^d : \|\mathbf{k}\|_2 \leq \|\mathbf{k} - \mathbf{G}\|_2 \text{ for all } \mathbf{G} \in \mathcal{R}^* \setminus \{\mathbf{0}\} \right\}. \tag{1.4}$$

For the standard cubic lattice $\mathcal{R} = \mathbb{Z}^d$, $\mathcal{B} = [-\pi, \pi]^d$.

1.2 Bloch decomposition of a periodic Hamiltonian

We work in a tight-binding setting: the Hamiltonian H is a self-adjoint operator on the Hilbert space $\mathcal{H} = \ell^2(\mathcal{R}, \mathbb{C}^M)$, where $M \geq 1$ is the number of internal degrees of freedom (typically orbitals) attached to each lattice site. The operator is described by a family of matrices $(H_\mathbf{R})_{\mathbf{R}\in\mathcal{R}}$ in $\mathbb{C}^{M\times M}$, encoding hopping amplitudes between sites separated by $\mathbf{R}$, with $H_{-\mathbf{R}} = H^*_\mathbf{R}$ (to ensure $H(\mathbf{k})$ is Hermitian).

[*]POEMS, CNRS, Inria, ENSTA, Institut Polytechnique de Paris, 91120 Palaiseau, France
[†]Université Paris-Saclay, CNRS, Laboratoire de mathématiques d'Orsay, 91405, Orsay, France

Periodicity allows us to apply the *Bloch transform*, which diagonalises translations and reduces the spectral problem for H on an infinite-dimensional space to a parametric family of finite-dimensional spectral problems indexed by $\mathbf{k} \in \mathcal{B}$. For each $\mathbf{k} \in \mathcal{B}$, define

$$H(\mathbf{k}) = \sum_{\mathbf{R} \in \mathcal{R}} \mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}} H_{\mathbf{R}} \in \mathbb{C}^{M\times M}. \tag{1.5}$$

In practice the sum is finite, so $H(\mathbf{k})$ is a trigonometric polynomial in $\mathbf{k}$, hence an entire analytic function of $\mathbf{k} \in \mathbb{C}^d$, and is $\mathcal{R}^*$-periodic. We denote its eigenvalues, ordered increasingly, by

$$\varepsilon_1(\mathbf{k}) \leq \varepsilon_2(\mathbf{k}) \leq \cdots \leq \varepsilon_M(\mathbf{k}), \tag{1.6}$$

and refer to the maps $\mathbf{k} \mapsto \varepsilon_n(\mathbf{k})$ as the *energy bands*. Each ε_n is analytic and $\mathcal{R}^*$-periodic, but generally only piecewise smooth: smoothness is lost at *band crossings*, i.e. at points $\mathbf{k}$ where two eigenvalues coincide.

Remark. *The tight-binding setting is not restrictive in practice. Continuous Schrödinger operators on $\mathbb{R}^d$ with $\mathcal{R}$-periodic potentials admit a similar Bloch decomposition. Truncating the resulting infinite spectrum to the bands relevant for the energy range of interest (e.g. via Wannier interpolation [***yates2007spectral***]) yields a finite-dimensional model of the form* (1.5)*.*

1.3 Supercell discretisation and the Monkhorst-Pack grid

Pas forcément utile dans le papier mais c'est important de comprendre comment les intégrales de Brillouin sont calculées en pratique.

For numerical purposes, the Brillouin zone is sampled on a uniform grid. This is most easily understood through the *supercell* construction. For $L \in \mathbb{N}^*$, the supercell is the L^d-fold replication of the unit cell along each lattice direction:

$$\Gamma_L = L\Gamma, \mathcal{R}_L = \mathcal{R} \cap \Gamma_L. \tag{1.7}$$

The truncated Bloch transform of a function $f \in L^2(\Gamma_L)$ reads

$$f(\mathbf{k}) = \sum_{\mathbf{R} \in \mathcal{R}_L} \mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}} f(\mathbf{R}), \tag{1.8}$$

and is evaluated on the dual of Γ_L, namely the *Monkhorst-Pack grid* [**monkhorst1976special**] (a uniform grid of L^d points in $\mathcal{B}$):

$$\mathcal{B}_L = \left\{ \sum_{i=1}^{d} \frac{n_i}{L} \mathbf{b}_i : n_i \in \{-\lfloor L/2 \rfloor, \ldots, \lfloor L/2 \rfloor - 1\} \right\}, \tag{1.9}$$

Brillouin-zone integrals are then approximated by the average

$$\frac{1}{|\mathcal{B}|} \int_{\mathcal{B}} f(\mathbf{k}) \mathrm{d}\mathbf{k} \approx \frac{1}{L^d} \sum_{\mathbf{k} \in \mathcal{B}_L} f(\mathbf{k}). \tag{1.10}$$

This is the periodic trapezoidal rule on $\mathcal{B}$ viewed as a torus. For analytic $\mathcal{R}^*$-periodic integrands, it converges exponentially fast in L [**trefethen2014exponentially**].

1.4 The density of states

The *density of states* (DOS) of H at energy $E \in \mathbb{R}$ is formally defined as

$$D(E) = \frac{1}{|\mathcal{B}|} \sum_{n=1}^{M} \int_{\mathcal{B}} \delta(E - \varepsilon_n(\mathbf{k})) \mathrm{d}\mathbf{k}. \tag{1.11}$$

It is a non-negative measure on $\mathbb{R}$, and $D(E)\mathrm{d}E$ counts the number of energy bands taking values in the infinitesimal interval $[E, E + \mathrm{d}E]$.

The expression (1.11) is delicate to handle numerically because the integrand is a Dirac measure. When the gradient $\nabla\varepsilon_n$ does not vanish on $\{\varepsilon_n = E\}$, the co-area formula [**cances2020numerical**] reformulates $D(E)$ as a sum of surface integrals:

$$D(E) = \frac{1}{|\mathcal{B}|} \sum_{n=1}^{M} \int_{S_n(E)} \frac{1}{|\nabla\varepsilon_n(\mathbf{k})|} \mathrm{d}\mathcal{H}^{d-1}(\mathbf{k}), \tag{1.12}$$

where $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional Hausdorff measure and

$$S_n(E) = \{\mathbf{k} \in \mathcal{B} : \varepsilon_n(\mathbf{k}) = E\} \tag{1.13}$$

is the n-th *Fermi surface* at energy E. The energies at which the non-vanishing-gradient condition fails (or at which two bands cross at energy E) are called *van Hove singularities*; they form a set of measure zero and are responsible for the only points where D is singular [**vanhove1953occurrence**]. In 1D the DOS diverges as $|E - \varepsilon_n(k_0)|^{-1/2}$ at a van Hove singularity k_0, while in 2D it typically diverges as $|\log|E - \varepsilon_n(\mathbf{k}_0)||$ at an isolated singularity $\mathbf{k}_0$ and as $|E - \varepsilon_n(\mathbf{k}_0)|^{-1/2}$ along a singular curve of points $\mathbf{k}_0$. In 3D, the DOS is typically continuous at van Hove singularities, but exhibits cusp-like features.

Equation (1.12) is the starting point of our numerical method: rather than regularising the Dirac in (1.11), we discretise the right-hand side directly, using a high-order quadrature on implicitly defined surfaces.

2 High-order quadrature on implicitly defined surfaces

We now recall the high-order quadrature method of Saye [**saye2015highorder**] for the integration of smooth functions over surfaces and volumes defined implicitly inside a hyperrectangle. Specialising it to $\phi = \varepsilon_n - E$ will yield a quadrature for the DOS via (1.12).

2.1 Setting

Let $U = (x_1^L, x_1^U) \times \cdots \times (x_d^L, x_d^U) \subset \mathbb{R}^d$ be a hyperrectangle and $\phi \in \mathcal{C}^k(U)$ a level-set function (with k at least equal to the desired order of accuracy). We seek to evaluate

$$\int_{\Gamma\cap U} g \mathrm{d}\mathcal{H}^{d-1}, \Gamma = \{\mathbf{x} \in \mathbb{R}^d : \phi(\mathbf{x}) = 0\}, \tag{2.1}$$

for a smooth integrand g. We denote by $(\mathbf{e}_i)_{i=1}^d$ the canonical basis of $\mathbb{R}^d$.

2.2 Dimension reduction via the implicit function theorem

The key idea is to express $\Gamma \cap U$ locally as the graph of an implicit *height function* over a $(d-1)$-dimensional sub-hyperrectangle, thereby reducing the integration on a $(d-1)$-dimensional surface in $\mathbb{R}^d$ to an integration on a $(d-1)$-dimensional volume in $\mathbb{R}^{d-1}$.

Suppose we can find a coordinate direction $i \in \{1, \ldots, d\}$ along which ϕ is monotone on U, in the sense that $\partial_i \phi$ is bounded away from zero on U. By the implicit function theorem, the equation $\phi(\mathbf{x}) = 0$ then defines a unique smooth height function $h : \tilde{U} \to (x_i^L, x_i^U)$, where $\tilde{U} \subset \mathbb{R}^{d-1}$ is the projection of U onto the coordinates $\tilde{\mathbf{x}} = (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_d)$, satisfying

$$\phi\left(\tilde{\mathbf{x}} + h(\tilde{\mathbf{x}})\mathbf{e}_i\right) = 0, \tilde{\mathbf{x}} \in \tilde{U}. \tag{2.2}$$

The subdomain $\tilde{V} \subset \tilde{U}$ on which the graph $\tilde{\mathbf{x}} \mapsto \tilde{\mathbf{x}} + h(\tilde{\mathbf{x}})\mathbf{e}_i$ indeed lies in U is determined by the signs of the restrictions of ϕ to the lower and upper faces of U in the direction $\mathbf{e}_i$:

$$\psi^L(\tilde{\mathbf{x}}) = \phi(\tilde{\mathbf{x}} + x_i^L \mathbf{e}_i), \psi^U(\tilde{\mathbf{x}}) = \phi(\tilde{\mathbf{x}} + x_i^U \mathbf{e}_i). \tag{2.3}$$

For instance, when $\partial_i \phi > 0$ on U, one has $\tilde{V} = \{\psi^L < 0\} \cap \{\psi^U > 0\}$. The other case is symmetric. All configurations are illustrated in Figure 1.

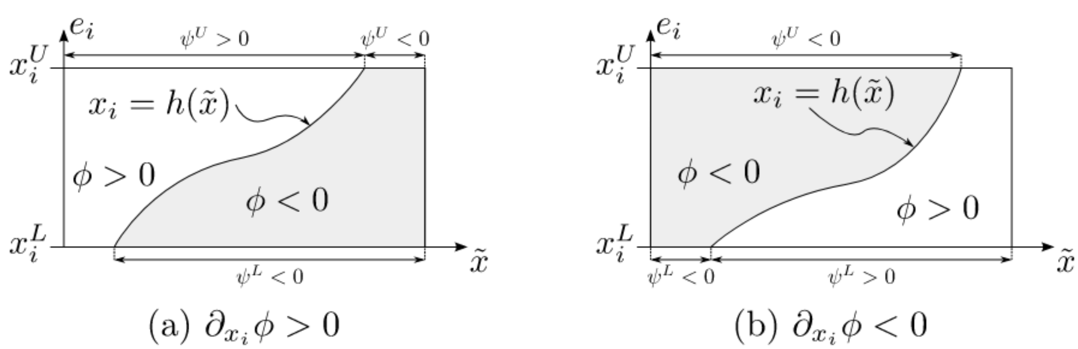

Figure 1: Cases for the definition of V_1 and V_2 (image from [**saye2015highorder**]).

A short computation using (2.2) gives the area element on Γ:

$$\sqrt{1 + \|\nabla_{\tilde{\mathbf{x}}} h\|^2} = \left.\frac{|\nabla \phi|}{|\partial_i \phi|}\right|_{\mathbf{x} = \tilde{\mathbf{x}} + h(\tilde{\mathbf{x}})\mathbf{e}_i}, \tag{2.4}$$

so that

$$\int_{\Gamma \cap U} g \mathrm{d}\mathcal{H}^{d-1} = \int_{\tilde{V}} g\left(\tilde{\mathbf{x}} + h(\tilde{\mathbf{x}})\mathbf{e}_i\right) \left.\frac{|\nabla \phi|}{|\partial_i \phi|}\right|_{\mathbf{x} = \tilde{\mathbf{x}} + h(\tilde{\mathbf{x}})\mathbf{e}_i} \mathrm{d}\tilde{\mathbf{x}}. \tag{2.5}$$

The $(d-1)$-dimensional integral on the right-hand side can itself be reduced to a $(d-2)$-dimensional one by applying the same procedure recursively, until a one-dimensional quadrature is reached. The final scheme is a tensor-product Gauss-Legendre quadrature in which the height function h is evaluated by a one-dimensional root-finding procedure for ϕ.

2.3 Validity and the role of guaranteed bounds

For the dimension-reduction step (2.5) to be valid, two ingredients must be checked on U:

1. there exists a coordinate direction i along which $\partial_i\phi$ has constant, known sign on U;
2. the signs of ψ^L and ψ^U on $\tilde{U}$ are correctly identified, so that $\tilde{V}$ matches the true projection of $\Gamma \cap U$.

Both checks reduce to verifying inequalities of the form "ϕ (or $\partial_i\phi$) does not change sign on a hyperrectangle". Saye's algorithm performs them by computing *guaranteed bounds* on ϕ and $\partial_i\phi$ over U. If a sign cannot be ascertained on U, the box is bisected and the procedure recurses on the sub-boxes. The overall convergence therefore hinges entirely on the quality of the available bounds on ϕ and its first partial derivatives.

2.4 Application to the density of states

For the DOS, we apply this procedure to each band n and to the level-set function

$$\phi_{n,E}(\mathbf{k}) = \varepsilon_n(\mathbf{k}) - E. \tag{2.6}$$

Combined with (1.12), this yields a quadrature for $D(E)$ as a sum, over a hierarchical subdivision of $\mathcal{B}$, of integrals of $g = 1/|\nabla\varepsilon_n|$ over the local pieces of $S_n(E)$.

The boxes produced by the subdivision are centered hyperrectangles

$$D = \mathbf{k}_0 + \boldsymbol{\delta}\mathbf{k} = \{\mathbf{k} \in \mathcal{B} : |k_i - k_{0,i}| \leq \delta k_i,\ i = 1, \dots, d\}, \tag{2.7}$$

and the bounds required by Saye's algorithm are bounds on $\varepsilon_n - E$, $\partial_i\varepsilon_n$ and (for the integrand $1/|\nabla\varepsilon_n|$) on the second derivatives $\partial_i\partial_j\varepsilon_n$, uniformly on each such box D.

3 Theoretical preliminaries

We collect in this section the analytic tools needed to construct, in Section 4, computable bounds on the Bloch eigenvalues and their derivatives.

Throughout, $H : \mathcal{B} \to \mathbb{C}^{M\times M}$ denotes an analytic Bloch Hamiltonian at som generic point $\mathbf{k}$ of $\mathcal{B}$. ε_n, ψ_n denote its eigenvalues and normalised eigenvectors at the same point $\mathbf{k}$.

3.1 Matrix norms

We use two matrix norms on $\mathbb{C}^{M\times M}$. The *operator (spectral) norm* is the induced norm associated with the Euclidean norm on $\mathbb{C}^M$:

$$\|A\|_{\mathrm{op}} = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^*A)}. \tag{3.1}$$

The *Frobenius norm* is

$$\|A\|_F = \sqrt{\sum_{i,j=1}^{M} |A_{ij}|^2} = \sqrt{\sum_{i=1}^{M} \sigma_i(A)^2}. \tag{3.2}$$

For Hermitian H, the spectral norm coincides with the spectral radius:

$$\|H\|_{\mathrm{op}} = \max_{\|x\|_2=1} \langle x|H|x\rangle = \max_{1\le i\le M} |\lambda_i(H)|. \tag{3.3}$$

The two norms are related by the elementary inequality

$$\|A\|_{\mathrm{op}} \le \|A\|_F \le \sqrt{M}\|A\|_{\mathrm{op}}. \tag{3.4}$$

Remark. *For Hermitian matrices, the induced* 1*-norm and* ∞*-norm coincide, but they are not unitarily invariant and are therefore ill-suited for spectral perturbation results; we do not use them in this paper. The spectral norm is natural for eigenvalue perturbation estimates, yet it is numerically difficult to bound over a set of parameters without solving a global optimization problem. The Frobenius norm, in contrast, is easily computable from the matrix entries and can be used to bound parameter-dependent spectral norms via* (3.4)*, albeit at the cost of a factor* $\sqrt{M}$*.*

3.2 Weyl's inequality

The starting point of all our eigenvalue bounds is Weyl's inequality, which controls the perturbation of individual eigenvalues by the spectral norm of the perturbation. We state it directly in the form best suited to the $\mathbf{k}$-dependence of the Bloch Hamiltonian.

Lemma 3.1 (Weyl's inequality for Bloch eigenvalues)**.** *For all* $n \in \{1, \dots, M\}$ *and all* $\mathbf{k}, \mathbf{k}_0 \in \mathcal{B}$*,*

$$|\varepsilon_n(\mathbf{k}) - \varepsilon_n(\mathbf{k}_0)| \le \|H(\mathbf{k}) - H(\mathbf{k}_0)\|_{\mathrm{op}}. \tag{3.5}$$

This inequality holds without any non-degeneracy assumption and relies only on the continuity of H. Its strength is to reduce the analysis of the (generally non-smooth) eigenvalue maps ε_n to the analysis of the (entire and analytic) matrix-valued H.

3.3 Time-independent perturbation theory: non-degenerate case

Weyl's inequality controls eigenvalue *values* but says nothing about their *derivatives*, which we need to control the integrand $1/|\nabla\varepsilon_n|$ of (1.12) and to certify the applicability of the implicit function theorem in Saye's algorithm. To access derivatives, we use standard time-independent perturbation theory, which expands the eigenvalues and eigenvectors of a perturbed self-adjoint operator in powers of the perturbation parameter.

3.3.1 General setup

Consider three Hermitian matrices $H_0, \delta H, H \in \mathbb{C}^{M\times M}$ related by

$$H(\lambda) = H_0 + \lambda\delta H, \tag{3.6}$$

where $\lambda \in \mathbb{R}$ is a small parameter. Let $\varepsilon_{n,0}, \psi_{n,0}$ denote the eigenpairs of H_0, assumed non-degenerate, and let $\varepsilon_n(\lambda), \psi_n(\lambda)$ denote those of $H(\lambda)$, depending analytically on λ for λ sufficiently small. Expanding

$$\varepsilon_n(\lambda) = \sum_{i=0}^{+\infty} \lambda^i \varepsilon_n^{(i)} \tag{3.7}$$

and identifying coefficients term by term, we obtain up to second order [**kato1995perturbation**],

$$\varepsilon_n(\lambda) = \varepsilon_{n,0} + \lambda\langle\psi_{n,0}|\delta H|\psi_{n,0}\rangle + \lambda^2 \sum_{m\neq n} \frac{|\langle\psi_{m,0}|\delta H|\psi_{n,0}\rangle|^2}{\varepsilon_{n,0} - \varepsilon_{m,0}} + \mathcal{O}(\lambda^3). \tag{3.8}$$

The first-order coefficient is the projection of δH onto the eigenstate $\psi_{n,0}$; the second-order coefficient involves all other eigenstates and is inversely proportional to the energy gaps separating $\varepsilon_{n,0}$ from the rest of the spectrum.

3.3.2 Application to the Bloch Hamiltonian

We now specialise this expansion to perturbations in $\mathbf{k}$. Fix $\mathbf{k}_0 \in \mathcal{B}$ and consider the Taylor expansion (as H is analytic)

$$H(\mathbf{k}) = H(\mathbf{k}_0) + \sum_{i=1}^{d} \delta k_i \partial_i H(\mathbf{k}_0) + \frac{1}{2}\sum_{i,j=1}^{d} \delta k_i \delta k_j \partial_i\partial_j H(\mathbf{k}_0) + \mathcal{O}(\|\boldsymbol{\delta}\mathbf{k}\|^3), \tag{3.9}$$

with $\delta k_i = k_i - k_{0,i}$. Plugging this into (3.8), one recovers the standard expressions for the Taylor expansion of ε_n.

$$\varepsilon_n(\mathbf{k}) = \varepsilon_n(\mathbf{k}_0) + \sum_{i=1}^{d} \delta k_i \partial_i \varepsilon_n(\mathbf{k}_0) + \frac{1}{2}\sum_{i,j=1}^{d} \delta k_i \delta k_j \partial_i\partial_j \varepsilon_n(\mathbf{k}_0) + \mathcal{O}(\|\boldsymbol{\delta}\mathbf{k}\|^3), \tag{3.10}$$

Writing $\psi_n = \psi_n(\mathbf{k}_0)$ for brevity, the first derivative is given by the Hellmann-Feynman formula

$$\partial_i \varepsilon_n(\mathbf{k}_0) = \langle\psi_n|\partial_i H(\mathbf{k}_0)|\psi_n\rangle, \tag{3.11}$$

and the second derivative is

$$\partial_i\partial_j \varepsilon_n(\mathbf{k}_0) = \langle\psi_n|\partial_i\partial_j H(\mathbf{k}_0)|\psi_n\rangle + 2\sum_{m\neq n} \frac{\mathrm{Re}\left(\langle\psi_n|\partial_i H(\mathbf{k}_0)|\psi_m\rangle\langle\psi_m|\partial_j H(\mathbf{k}_0)|\psi_n\rangle\right)}{\varepsilon_n(\mathbf{k}_0) - \varepsilon_m(\mathbf{k}_0)}. \tag{3.12}$$

The denominator $\varepsilon_n - \varepsilon_m$ in (3.12) encodes the influence of the rest of the spectrum on the curvature of the band ε_n. It also indicates the regime in which the expansion fails: when another band approaches ε_n, the second derivative diverges and the eigenvalue ceases to be smooth.

4 Bounds for eigenvalues and derivatives

We now combine the analytic tools of Section 3 into computable bounds on the Bloch eigenvalues and their derivatives. All bounds are established on a fixed centered box

$$D = \mathbf{k}_0 + \boldsymbol{\delta}\mathbf{k} = \{\mathbf{k} \in \mathcal{B} : |k_i - k_{0,i}| \leq \delta k_i,\ i = 1, \dots, d\}, \tag{4.1}$$

issued from the adaptive subdivision performed by Saye's algorithm. Throughout, the half-widths $\delta k_i \geq 0$ are non-negative scalars. We adopt the convention that the path bounds are first stated in operator norm — the natural norm for spectral perturbation arguments — and only later substituted by Frobenius norms for numerical evaluation (see Section 5).

4.1 The path bound

The starting point of all our eigenvalue bounds is the following pathwise estimate, which controls the variation of the Hamiltonian on D by the first derivatives of H. The result is stated as a chain of two inequalities of decreasing sharpness, corresponding to two ways of treating the linear combination $\sum_j \partial_j H(\mathbf{q})\,(k_j - k_{0,j})$: either keeping the matrix absolute value as a single object (sharper, but requires a spectral decomposition to evaluate) or splitting it into individual operator norms (looser, but amenable to a Frobenius substitution).

Lemma 4.1 (First-order path bound)**.** *For all* $\mathbf{k} \in D$*,*

$$\|H(\mathbf{k}) - H(\mathbf{k}_0)\|_{\mathrm{op}} \le \sup_{\mathbf{q}\in D} \left\| \sum_{j=1}^{d} |\partial_j H(\mathbf{q})|\, \delta k_j \right\|_{\mathrm{op}} \tag{4.2}$$

$$\le \sup_{\mathbf{q}\in D} \sum_{j=1}^{d} \|\partial_j H(\mathbf{q})\|_{\mathrm{op}}\, \delta k_j. \tag{4.3}$$

Proof. The proof proceeds in two steps, one per inequality.

Step 1 (path bound, (4.2)). Since D is convex (it's a rectangle), the segment $\gamma : t \in [0,1] \mapsto (1-t)\mathbf{k}_0 + t\mathbf{k}$ lies in D. The map $t \mapsto H(\gamma(t))$ is analytic, and the chain rule gives $\frac{\mathrm{d}}{\mathrm{d}t} H(\gamma(t)) = \sum_{j=1}^{d} \partial_j H(\gamma(t))\,(k_j - k_{0,j})$. Integrating from 0 to 1,

$$H(\mathbf{k}) - H(\mathbf{k}_0) = \int_0^1 \sum_{j=1}^{d} \partial_j H(\gamma(t))\,(k_j - k_{0,j})\,\mathrm{d}t. \tag{4.4}$$

Fix $t \in [0,1]$ and set $B(t) := \sum_j \partial_j H(\gamma(t))\,(k_j - k_{0,j})$. The matrix $B(t)$ is Hermitian as a real linear combination of Hermitian matrices. We now establish the pointwise inequality

$$\|B(t)\|_{\mathrm{op}} \le \left\| \sum_{j=1}^{d} |\partial_j H(\gamma(t))|\, |k_j - k_{0,j}| \right\|_{\mathrm{op}}, \tag{4.5}$$

which is the heart of the argument. For any Hermitian matrix A with spectral decomposition $A = \sum_i \mu_i P_i$ and any unit vector v,

$$|\langle v, Av\rangle| = \left| \sum_i \mu_i \,\|P_i v\|^2 \right| \le \sum_i |\mu_i|\, \|P_i v\|^2 = \langle v, |A|\, v\rangle, \tag{4.6}$$

Combining (4.6) with the variational principle $\|B(t)\|_{\mathrm{op}} = \max_{\|v\|=1} |\langle v, B(t)v\rangle|$ (valid for Hermitian

$B(t)$),

$$\|B(t)\|_{\mathrm{op}} = \max_{\|v\|=1} \left| \sum_j (k_j - k_{0,j}) \langle v, \partial_j H(\gamma(t))\, v\rangle \right| \tag{4.7}$$

$$\le \max_{\|v\|=1} \sum_j |k_j - k_{0,j}| \, |\langle v, \partial_j H(\gamma(t))\, v\rangle| \tag{4.8}$$

$$\le \max_{\|v\|=1} \sum_j |k_j - k_{0,j}| \, \langle v, |\partial_j H(\gamma(t))|\, v\rangle \tag{4.9}$$

$$= \left\| \sum_j |\partial_j H(\gamma(t))| \, |k_j - k_{0,j}| \right\|_{\mathrm{op}}, \tag{4.10}$$

where the last equality uses that $\sum_j |\partial_j H(\gamma(t))| \, |k_j - k_{0,j}|$ is positive semi-definite (a non-negative combination of PSD matrices), so its operator norm coincides with its largest eigenvalue. This proves (4.5).

Taking the operator norm of (4.4) via the Bochner integral triangle inequality and applying (4.5),

$$\|H(\mathbf{k}) - H(\mathbf{k}_0)\|_{\mathrm{op}} \le \int_0^1 \|B(t)\|_{\mathrm{op}} \, \mathrm{d}t \le \int_0^1 \left\| \sum_j |\partial_j H(\gamma(t))| \, |k_j - k_{0,j}| \right\|_{\mathrm{op}} \mathrm{d}t. \tag{4.11}$$

To replace the path-dependent integrand by a uniform bound on D, we use $|k_j - k_{0,j}| \le \delta k_j$ and the monotonicity of the operator norm for positive semi-deff positive seminite matrices: if $0 \le P \le Q$ (in the PSD sense) then $\|P\|_{\mathrm{op}} \le \|Q\|_{\mathrm{op}}$. Applied to

$$\sum_j |\partial_j H(\gamma(t))| \, |k_j - k_{0,j}| \le \sum_j |\partial_j H(\gamma(t))| \, \delta k_j, \tag{4.12}$$

this gives

$$\left\| \sum_j |\partial_j H(\gamma(t))| \, |k_j - k_{0,j}| \right\|_{\mathrm{op}} \le \left\| \sum_j |\partial_j H(\gamma(t))| \, \delta k_j \right\|_{\mathrm{op}} \le \sup_{\mathbf{q}\in D} \left\| \sum_j |\partial_j H(\mathbf{q})| \, \delta k_j \right\|_{\mathrm{op}}, \tag{4.13}$$

where the last inequality uses $\gamma(t) \in D$. The right-hand side is constant in t, so integrating over $[0, 1]$ yields (4.2).

Since $\||\partial_j H(\mathbf{q})|\|_{\mathrm{op}} = \|\partial_j H(\mathbf{q})\|_{\mathrm{op}}$ (the operator norm is invariant under taking the matrix absolute value, as it depends only on singular values), applying the triangular inequality yields (4.3). □

Remark (Sharpness: (4.2) versus (4.3))**.** *The bound (4.2) can be strictly tighter than (4.3), with the gap controlled by the spectral overlap of the matrices $(|\partial_j H(\mathbf{q})|)_j$. Two extreme cases illustrate the spread.*

- *If all $|\partial_j H(\mathbf{q})|$ share an eigenvector v achieving their largest eigenvalue, then equality holds: $\| \sum_j |\partial_j H(\mathbf{q})| \, \delta k_j\|_{\mathrm{op}} = \sum_j \|\partial_j H(\mathbf{q})\|_{\mathrm{op}} \, \delta k_j$.*

- *If the $|\partial_j H(\mathbf{q})|$ are mutually orthogonal projectors (e.g. $|A_1| = |e_1\rangle\langle e_1|$, $|A_2| = |e_2\rangle\langle e_2|$), then $\|\sum_j |\partial_j H|\,\delta k_j\|_{\mathrm{op}} = \max_j \delta k_j$, whereas $\sum_j \|\partial_j H\|_{\mathrm{op}}\,\delta k_j = \sum_j \delta k_j$, so the second bound is up to d times looser.*

A naive Cauchy-Schwarz bound $|\langle v, \partial_j H\, v\rangle| \le \|\partial_j H\|_{\mathrm{op}}$ directly applied to $\|\sum_j (k_j - k_{0,j})\partial_j H\|_{\mathrm{op}}$ gives the looser bound (4.3) only; the spectral inequality (4.6) is what allows (4.2).

Remark (Matrix absolute value and spectral decomposition)**.** *The matrix absolute value $|A|$ of a Hermitian matrix A is defined as $|A| = \sqrt{A^2}$. It can be computed from the spectral decomposition of A: if $A = \sum_i \mu_i P_i$ then $|A| = \sum_i |\mu_i| P_i$. The bound using $|\partial_j H|$ is therefore sharper than the one using $\|\partial_j H\|_{\mathrm{op}}$ because it retains the spectral information of $\partial_j H$ instead of discarding it via the operator norm. However, it is not computable on an arbitrary set D without solving a global optimization problem, since the spectral decomposition of $\sum_j |\partial_j H(\mathbf{q})|\,\delta k_j$ depends on $\mathbf{q}$. The bound using $\|\partial_j H\|_{\mathrm{op}}$ is therefore more tractable.*

4.2 First-order eigenvalue bounds

Combining Weyl's inequality (Lemma 3.1) with the path bound (4.2)–(4.3) yields the order-one bound on the eigenvalues, valid uniformly in n and without any non-degeneracy assumption.

Corollary 4.1 (First-order eigenvalue bound)**.** *For all $n \in \{1, \dots, M\}$ and all $\mathbf{k} \in D$,*

$$|\varepsilon_n(\mathbf{k}) - \varepsilon_n(\mathbf{k}_0)| \le \sup_{\mathbf{q}\in D} \left\| \sum_{j=1}^{d} |\partial_j H(\mathbf{q})|\,\delta k_j \right\|_{\mathrm{op}} \tag{4.14}$$

$$\le \sup_{\mathbf{q}\in D} \sum_{j=1}^{d} \|\partial_j H(\mathbf{q})\|_{\mathrm{op}}\,\delta k_j. \tag{4.15}$$

Proof. Direct application of Lemma 3.1 and Lemma 4.1. □

Remark (Range of validity)**.** *Both bounds are valid uniformly in n and require no assumption on the multiplicity or isolation of $\varepsilon_n(\mathbf{k}_0)$, since Weyl's inequality and the path bound both hold for every hermitian Hamiltonian. They are therefore the appropriate bounds for boxes that may contain band crossings, where the order-two bound below fails.*

4.3 Second-order eigenvalue bound

A sharper bound is obtained by pushing the Taylor expansion of ε_n to second order, at the price of requiring a spectral gap.

Lemma 4.2 (Second-derivative bound)**.** *Assume that the band ε_n is non-degenerate on D, i.e. $\Delta_n(D) > 0$, with*

$$\Delta_n(D) := \inf_{\mathbf{k}\in D} \min_{m\ne n} |\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})|. \tag{4.16}$$

Then for all $\mathbf{q} \in D$ and all $i, j \in \{1, \dots, d\}$,

$$|\partial_i\partial_j \varepsilon_n(\mathbf{q})| \le \mu_{ij}(\mathbf{q}) := \|\partial_i\partial_j H(\mathbf{q})\|_{\mathrm{op}} + \frac{2\|\partial_i H(\mathbf{q})\|_{\mathrm{op}}\,\|\partial_j H(\mathbf{q})\|_{\mathrm{op}}}{\Delta_n(D)}. \tag{4.17}$$

Proof. Under the non-degeneracy assumption, ε_n is real-analytic on D, and the second-derivative formula (3.12) applies. The triangle inequality decomposes $|\partial_i \partial_j \varepsilon_n(\mathbf{q})|$ into two terms.

The first term is bounded as

$$|\langle \psi_n | \partial_i \partial_j H(\mathbf{q}) | \psi_n \rangle| \le \|\partial_i \partial_j H(\mathbf{q})\|_{\mathrm{op}}, \tag{4.18}$$

since $\|\psi_n\|_2 = 1$.

For the second term, the gap denominator is bounded below by $\Delta_n(D)$, and the numerator is controlled via the Cauchy–Schwarz inequality:

$$\begin{aligned}
\left| \sum_{m \ne n} \langle \psi_n | A | \psi_m \rangle \langle \psi_m | B | \psi_n \rangle \right| &\le \sum_{m \ne n} |\langle \psi_n | A | \psi_m \rangle| \, |\langle \psi_m | B | \psi_n \rangle| && (4.19) \\
&\le \left(\sum_{m \ne n} |\langle \psi_n | A | \psi_m \rangle|^2 \right)^{1/2} \left(\sum_{m \ne n} |\langle \psi_m | B | \psi_n \rangle|^2 \right)^{1/2} && (4.20) \\
&\le \|A\psi_n\|_2 \, \|B^* \psi_n\|_2 \le \|A\|_{\mathrm{op}} \, \|B\|_{\mathrm{op}}, && (4.21)
\end{aligned}$$

where we used that $\sum_{m \ne n} |\langle \psi_n | A | \psi_m \rangle|^2 \le \sum_m |\langle \psi_n | A | \psi_m \rangle|^2 = \|A\psi_n\|_2^2$. Setting $A = \partial_i H(\mathbf{q})$ and $B = \partial_j H(\mathbf{q})$ yields the second term in (4.17), and the result follows. □

Corollary 4.2 (Second-order eigenvalue bound)**.** *Under the assumptions of Lemma 4.2, for all* $\mathbf{k} \in D$*,*

$$\left| \varepsilon_n(\mathbf{k}) - \varepsilon_n(\mathbf{k}_0) - \sum_{i=1}^{d} \partial_i \varepsilon_n(\mathbf{k}_0) \, (k_i - k_{0,i}) \right| \le \frac{1}{2} \sup_{\mathbf{q} \in D} \sum_{i,j=1}^{d} \mu_{ij}(\mathbf{q}) \, \delta k_i \, \delta k_j. \tag{4.22}$$

Proof. The non-degeneracy assumption guarantees that ε_n is $\mathcal{C}^2$ on D. By Taylor–Lagrange applied to there exists $\mathbf{q}$ on the segment $[\mathbf{k}_0, \mathbf{k}] \subset D$ such that

$$\varepsilon_n(\mathbf{k}) - \varepsilon_n(\mathbf{k}_0) - \sum_i \partial_i \varepsilon_n(\mathbf{k}_0) \, (k_i - k_{0,i}) = \frac{1}{2} \sum_{i,j} \partial_i \partial_j \varepsilon_n(\mathbf{q}) \, (k_i - k_{0,i})(k_j - k_{0,j}). \tag{4.23}$$

Bounding each $|\partial_i \partial_j \varepsilon_n(\mathbf{q})|$ by $\mu_{ij}(\mathbf{q})$ via Lemma 4.2 and using $|k_i - k_{0,i}| \le \delta k_i$ together with $\mathbf{q} \in D$ yields the claim. □

Remark (When does the order-two bound improve over the order-one bound?)**.** *Unlike Corollary 4.1, the order-two bound is not uniform in n: it depends on the gap $\Delta_n(D)$, and is therefore meaningful only on boxes where the band ε_n is isolated and the gap is not too small. Asymptotically as $\boldsymbol{\delta}\mathbf{k} \to 0$ at fixed gap, the right-hand side of (4.22) is $\mathcal{O}(\|\boldsymbol{\delta}\mathbf{k}\|^2)$, whereas the right-hand side of (4.14) controls $\varepsilon_n - \varepsilon_n(\mathbf{k}_0)$ at order $\mathcal{O}(\|\boldsymbol{\delta}\mathbf{k}\|)$. The order-two bound is therefore preferable on small enough boxes; the crossover scale is roughly $\|\boldsymbol{\delta}\mathbf{k}\| \sim \Delta_n(D)/\|\partial_i H\|_{\mathrm{op}}$, the inverse of the Hamiltonian Lipschitz constant relative to the gap.*

4.4 Bounds on the spectral gap

The order-two bound and the derivative bound below involve the inverse of the gap $1/\Delta_n(D)$ throughs second order terms. To make those bounds effective, we need a guaranteed strictly positive lower bound on $\Delta_n(D)$. Two such bounds follow as immediate corollaries of the eigenvalue bounds, and a sharper one is obtained from the order-two bound.

Corollary 4.3 (Lower bounds on the spectral gap)**.** *For any* $\mathbf{k} \in D$ *and any* $m \neq n$*,*

$$|\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})| \geq |\varepsilon_n(\mathbf{k}_0) - \varepsilon_m(\mathbf{k}_0)| - |\varepsilon_n(\mathbf{k}) - \varepsilon_n(\mathbf{k}_0)| - |\varepsilon_m(\mathbf{k}) - \varepsilon_m(\mathbf{k}_0)|. \tag{4.24}$$

Bounding the last two terms by Lemma 3.1 or by the path bound (4.3) yields, respectively,

$$\Delta_n(D) \geq \min_{m \neq n} \left(|\varepsilon_n(\mathbf{k}_0) - \varepsilon_m(\mathbf{k}_0)| - 2 \sup_{\mathbf{q} \in D} \|H(\mathbf{q}) - H(\mathbf{k}_0)\|_{\mathrm{op}} \right)_+, \tag{4.25}$$

$$\Delta_n(D) \geq \min_{m \neq n} \left(|\varepsilon_n(\mathbf{k}_0) - \varepsilon_m(\mathbf{k}_0)| - 2 \sup_{\mathbf{q} \in D} \sum_{j=1}^{d} \|\partial_j H(\mathbf{q})\|_{\mathrm{op}}\, \delta k_j \right)_+, \tag{4.26}$$

where $(x)_+ = \max(x, 0)$*. The same inequalities hold with* $\|\cdot\|_{\mathrm{op}}$ *replaced by the tighter bound (4.2).*

Remark. *Both bounds are informative only when their right-hand side is strictly positive, that is, when the band* ε_n *remains isolated from its neighbours on* D*. The non-positivity of (4.26) can signal near-degeneracies, and trigger a subdivision of* D *by Saye's algorithm.*

A still sharper gap bound can be derived from the order-two eigenvalue bound (4.22). Unlike the previous two bounds, this one is no longer uniform in n: the band of interest ε_n and a competing band ε_m generally have different remainders.

Corollary 4.4 (Order-two lower bound on the spectral gap)**.** *Writing* $L_n(\mathbf{k}) := \sum_i \partial_i \varepsilon_n(\mathbf{k}_0)\,(k_i - k_{0,i})$ *for the linear part and* $Q_n(D) := \frac{1}{2} \sup_{\mathbf{q} \in D} \sum_{i,j} \mu_{ij}(\mathbf{q})\, \delta k_i\, \delta k_j$ *for the quadratic remainder of band* n*, with* μ_{ij} *defined in (4.17),*

$$\Delta_n(D) \geq \min_{m \neq n} \inf_{\mathbf{k} \in D} \left(|\varepsilon_n(\mathbf{k}_0) - \varepsilon_m(\mathbf{k}_0) + L_n(\mathbf{k}) - L_m(\mathbf{k})| - Q_n(D) - Q_m(D) \right)_+. \tag{4.27}$$

The inner infimum of $L_n - L_m$ *over* $\mathbf{k} \in D$ *is explicit: being affine in* $\mathbf{k}$*, it is attained at a vertex of* D*.*

Proof. Apply Corollary 4.2 to bands n and m: $\varepsilon_l(\mathbf{k}) = \varepsilon_l(\mathbf{k}_0) + L_l(\mathbf{k}) + r_l(\mathbf{k})$ with $|r_l(\mathbf{k})| \leq R_l(D)$ for $l \in \{n, m\}$. Then

$$|\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})| \geq |\varepsilon_n(\mathbf{k}_0) - \varepsilon_m(\mathbf{k}_0) + L_n(\mathbf{k}) - L_m(\mathbf{k})| - |r_n(\mathbf{k})| - |r_m(\mathbf{k})|, \tag{4.28}$$

by the reverse triangle inequality. Bounding the last two terms by $Q_n(D)$ and $Q_m(D)$ and taking the infimum over $\mathbf{k} \in D$ and the minimum over $m \neq n$ yields the claim. ☐

Circular dependency. The quadratic remainders $Q_n(D), Q_m(D)$ depend through μ_{ij} on a positive lower bound on the spectral gap. To break this circularity, the order-two gap bound is built on top of the order-one and zero bounds: one first uses (4.26) to obtain a guaranteed lower bound on $\Delta_n(D)$, uses it to evaluate μ_{ij} and $Q_n(D)$, and then plugs the result into (4.27) to obtain a sharper estimate. The order-two gap bound therefore *refines* an existing certificate of non-degeneracy; it cannot create one.

Remark. *Numerically, the bounds are obtained using interval arithmetic.*

4.5 Bounds on eigenvalue derivatives

To certify the applicability of the implicit function theorem in Saye's algorithm, and to bound the integrand $1/|\nabla \varepsilon_n|$ of (1.12), we need uniform bounds on $\partial_i \varepsilon_n$ on D. They follow from Lemma 4.2 and a first-order Taylor expansion of $\partial_i \varepsilon_n$ in $\mathbf{k}$.

Corollary 4.5 (First-order bound on eigenvalue derivatives)**.** *Under the assumptions of Lemma 4.2, for all $\mathbf{k} \in D$ and all $i \in \{1, \ldots, d\}$,*

$$|\partial_i \varepsilon_n(\mathbf{k}) - \partial_i \varepsilon_n(\mathbf{k}_0)| \leq \sup_{\mathbf{q} \in D} \sum_{j=1}^{d} \mu_{ij}(\mathbf{q})\, \delta k_j, \tag{4.29}$$

with μ_{ij} defined in (4.17).

Proof. Apply Taylor-Lagrange at order 1 to $\partial_i \varepsilon_n$ (which is $\mathcal{C}^1$ on D under the non-degeneracy assumption): there exists $\boldsymbol{\xi} \in [\mathbf{k}_0, \mathbf{k}] \subset D$ such that

$$\partial_i \varepsilon_n(\mathbf{k}) - \partial_i \varepsilon_n(\mathbf{k}_0) = \sum_{j=1}^{d} \partial_i \partial_j \varepsilon_n(\boldsymbol{\xi})\, (k_j - k_{0,j}). \tag{4.30}$$

Bounding each $|\partial_i \partial_j \varepsilon_n(\boldsymbol{\xi})|$ by $\mu_{ij}(\boldsymbol{\xi})$ via Lemma 4.2 and using $|k_j - k_{0,j}| \leq \delta k_j$ with $\boldsymbol{\xi} \in D$ yields the claim. □

Remark. *The spectral gap $\Delta_n(D)$ appears in μ_{ij} and must be replaced by a computable lower bound from Corollary 4.3 or Corollary 4.4. The bound on $|\nabla \varepsilon_n|$ is informative only on boxes where this lower bound is strictly positive.*

4.6 Theoretical comparison of the bounds

We summarise the bounds derived above and their hierarchy.

	Sharpness	Non-degeneracy	Easiness to compute	Numerically Computable
(3.1)	$\mathcal{O}(\delta \mathbf{k})$	No	Uniform	Yes
(4.14)	$\mathcal{O}(\delta \mathbf{k})$	No	Uniform	No
(4.15)	$\mathcal{O}(\delta \mathbf{k})$	No	Uniform	Yes
(4.22)	$\mathcal{O}(\delta \mathbf{k}^2)$	Yes	Non-uniform	Yes

Table 1: Comparison of the eigenvalue bounds derived in Section 4.

5 Numerical evaluation of bounds

The bounds of Section 4 are still expressed as suprema, over the continuous box D, of operator norms of H and its derivatives. To make them effective, two operations must be replaced by computable proxies:

1. the operator norm $\|\cdot\|_{\mathrm{op}}$, which requires a spectral decomposition and does not propagate well through interval arithmetic;

Bound	Sharpness	Needs gap	Uniform in n	Computable
Weyl (Lemma 3.1)	$\mathcal{O}(\Vert\boldsymbol{\delta}\mathbf{k}\Vert)$	No	Yes	Yes
Tight order-one (4.14)	$\mathcal{O}(\Vert\boldsymbol{\delta}\mathbf{k}\Vert)$	No	Yes	No
Loose order-one (4.15)	$\mathcal{O}(\Vert\boldsymbol{\delta}\mathbf{k}\Vert)$	No	Yes	Yes
Order-two (4.22)	$\mathcal{O}(\Vert\boldsymbol{\delta}\mathbf{k}\Vert^2)$	Yes	No	Yes

Table 2: Comparison of the eigenvalue bounds of Section 4. *Sharpness*: leading order in the box half-width $\boldsymbol{\delta}\mathbf{k}$. *Needs gap*: whether the bound requires the non-degeneracy assumption $\Delta_n(D) > 0$. *Uniform in n*: whether the bound is independent of the band index. *Computable*: whether the bound can be evaluated in interval arithmetic on D, possibly after the Frobenius substitution of Subsection 5.1. The tight order-one bound resists the Frobenius substitution () and is therefore not computable in practice.

2. the supremum over $\mathbf{q} \in D$, which is a continuous optimisation problem.

We address each point in turn.

5.1 From operator norm to Frobenius norm

The Frobenius norm $\|A\|_F = (\sum_{ij} |A_{ij}|^2)^{1/2}$ is computed directly from the matrix entries of A and is therefore propagated through interval arithmetic by standard entrywise operations. The substitution rests on the elementary inequality (3.4),

$$\|A\|_{\mathrm{op}} \leq \|A\|_F \leq \sqrt{M}\,\|A\|_{\mathrm{op}}, \tag{5.1}$$

which trades a factor at most $\sqrt{M}$ for direct entrywise computability.

Applying the substitution to the loose order-one bound (4.15) yields the practical bound

$$|\varepsilon_n(\mathbf{k}) - \varepsilon_n(\mathbf{k}_0)| \leq \sup_{\mathbf{q}\in D} \sum_{j=1}^{d} \|\partial_j H(\mathbf{q})\|_F\, \delta k_j. \tag{5.2}$$

Likewise, the second-derivative bound (4.17) becomes

$$\mu^F_{ij}(\mathbf{q}) := \|\partial_i\partial_j H(\mathbf{q})\|_F + \frac{2\|\partial_i H(\mathbf{q})\|_F\,\|\partial_j H(\mathbf{q})\|_F}{\Delta_n(D)} \geq \mu_{ij}(\mathbf{q}), \tag{5.3}$$

and the order-two bound (4.22) and derivative bound Corollary 4.5 hold verbatim with μ_{ij} replaced by μ^F_{ij}.

Remark (Why not Frobenius-substitute the tight bound (4.14)?)**.** *Naively, one might attempt*

$$\left\| \sum_j |\partial_j H(\mathbf{q})|\,\delta k_j \right\|_{\mathrm{op}} \leq \left\| \sum_j |\partial_j H(\mathbf{q})|\,\delta k_j \right\|_F \tag{5.4}$$

however the absolute value of a hermitian matrix requires a diagonalisation, which is not stable under interval arithmetic. The bound (5.4) is therefore not computable, and we do not use it.

Remark (Frobenius pessimism in dimension M)**.** *The factor $\sqrt{M}$ is conservative: it is attained only when all singular values are equal. For a tight-binding model with M Wannier bands and physically motivated sparsity (most matrix entries small), the actual ratio $\|A\|_F/\|A\|_{\mathrm{op}}$ is typically much smaller than $\sqrt{M}$.*

5.2 Analytic bound from Fourier coefficients

The Bloch transform (1.5) expresses H as a finite trigonometric polynomial in $\mathbf{k}$. Differentiating term by term,

$$\partial^{\boldsymbol{\alpha}} H(\mathbf{k}) = \sum_{\mathbf{R}\in\mathcal{R}} (i\mathbf{R})^{\boldsymbol{\alpha}} \mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}} H_{\mathbf{R}}, \tag{5.5}$$

for any multi-index $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d)$, with $(\mathbf{R})^{\boldsymbol{\alpha}} := \prod_l R_l^{\alpha_l}$. The triangle inequality on the Frobenius norm, applied entrywise after using $|\mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}}| = 1$, yields the global bound

$$\sup_{\mathbf{k}\in\mathcal{B}} \|\partial^{\boldsymbol{\alpha}} H(\mathbf{k})\|_F \leq \sum_{\mathbf{R}\in\mathcal{R}} |\mathbf{R}^{\boldsymbol{\alpha}}| \, \|H_{\mathbf{R}}\|_F. \tag{5.6}$$

This bound is independent of the box D, uniform on the entire Brillouin zone, and *precomputable* from the matrix entries of the $(H_{\mathbf{R}})_{\mathbf{R}}$ at zero asymptotic cost. It directly furnishes upper bounds on $\|\partial_j H\|_F$ (taking $\boldsymbol{\alpha} = \mathbf{e}_j$) and $\|\partial_i\partial_j H\|_F$ (taking $\boldsymbol{\alpha} = \mathbf{e}_i + \mathbf{e}_j$).

Remark (Pessimism of the Fourier bound)**.** *The bound (5.6) discards all phase information from* $\mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}}$ *and is therefore typically loose, by a factor that grows with the range of the* $H_{\mathbf{R}}$*.*

5.3 Interval arithmetic

The Frobenius substitution of Subsection 5.1 reduces every quantity entering our bounds to an entrywise expression in H and its derivatives. Whenever such an expression must be evaluated as a supremum over a box D — including the Frobenius substitute of $\sup_{\mathbf{q}\in D} \|H(\mathbf{q}) - H(\mathbf{k}_0)\|_{\mathrm{op}}$ appearing in Weyl's inequality (Lemma 3.1) and the derivative norms $\sup_{\mathbf{q}\in D} \|\partial^{\boldsymbol{\alpha}} H(\mathbf{q})\|_F$ — we propagate guaranteed enclosures through the arithmetic by means of *interval arithmetic* [**moore2009introduction**]. We recall the elementary rules in our setting.

Real intervals. A real interval is a closed connected subset $[a,b] \subset \mathbb{R}$. Elementary operations are defined so that the result encloses every value attainable from inputs in the operand intervals:

$$[a,b] + [c,d] = [a+c,\ b+d], \tag{5.7}$$

$$[a,b] - [c,d] = [a-d,\ b-c], \tag{5.8}$$

$$[a,b]\cdot[c,d] = \big[\min(ac,ad,bc,bd),\ \max(ac,ad,bc,bd)\big]. \tag{5.9}$$

Trigonometric functions sin and cos are evaluated by locating the critical points (integer multiples of $\pi/2$) inside $[a,b]$: when none lie inside, the bounds reduce to the values at the endpoints; otherwise they are corrected to ± 1 accordingly.

Complex intervals. A complex interval is a Cartesian rectangle $[a,b] + i\,[c,d] \subset \mathbb{C}$. Addition, subtraction, and multiplication reduce to the real-interval rules applied to real and imaginary parts. The only nontrivial operation in our setting is the unit-modulus phase

$$\mathrm{e}^{i[a,b]} = \cos[a,b] + i\sin[a,b]. \tag{5.10}$$

Application to the Bloch transform. For $\mathbf{k} \in D$, each k_l is a real interval $[k_{0,l} - \delta k_l,\ k_{0,l} + \delta k_l]$. For each $\mathbf{R} \in \mathcal{R}$, the dot product $\mathbf{k} \cdot \mathbf{R} = \sum_l k_l R_l$ is then a real interval and $\mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}}$ is a complex interval. Term-by-term differentiation of (1.5) and summation give, for any multi-index $\boldsymbol{\alpha}$, an entrywise complex-interval matrix

$$\left[\partial^{\boldsymbol{\alpha}} H(D)\right]_{pq} := \sum_{\mathbf{R}\in\mathcal{R}} (i\mathbf{R})^{\boldsymbol{\alpha}}\, \mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}}\, [H_{\mathbf{R}}]_{pq}, \tag{5.11}$$

which by construction contains $[\partial^{\boldsymbol{\alpha}} H(\mathbf{q})]_{pq}$ for every $\mathbf{q} \in D$ and every p, q. The squared Frobenius norm is then bounded entrywise:

$$\sup_{\mathbf{q}\in D} \|\partial^{\boldsymbol{\alpha}} H(\mathbf{q})\|_F^2 \leq \sum_{p,q=1}^{M} \max_{z\in[\partial^{\boldsymbol{\alpha}} H(D)]_{pq}} |z|^2, \tag{5.12}$$

where each maximum is the squared distance from the origin to the farthest corner of the complex rectangle, computable in closed form. The same construction with $H(\mathbf{q}) - H(\mathbf{k}_0)$ at entry level (interval subtraction with the constant matrix $H(\mathbf{k}_0)$) gives the Frobenius enclosure that makes Weyl's inequality effective on D, and (3.4) converts it back to an operator-norm bound at the cost of an extra factor $\sqrt{M}$.

Remark (Comparison with the analytic Fourier bound)**.** *The Fourier bound* (5.6)*, obtained by the triangle inequality after using* $|\mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}}| = 1$*, is recovered from* (5.11) *when the phase* $\mathrm{e}^{i\mathbf{k}\cdot\mathbf{R}}$ *is enclosed in the unit disc independently of* D*. The full interval-arithmetic procedure retains the actual range of the phase on* D *and is therefore strictly tighter on small boxes, at the cost of one enclosure computation per* D*. The Fourier bound, precomputable and box-independent, remains useful at the early stages of the subdivision where its lack of adaptivity is offset by its zero per-box cost.*

5.3.1 Sampled bound with Lipschitz correction

An estimate can be obtained by sampling $\Phi : \mathbf{q} \mapsto \|\partial_j H(\mathbf{q})\|_F$ (or any other Frobenius-norm expression in H and its derivatives) on a finite grid in D and adding a Lipschitz correction.

Let $N \in \mathbb{N}^*$ and let $D_N \subset D$ be a uniform discretized D with N^d points such that any $\mathbf{q}^* \in D$ lies within distance $\delta k_j / N$ of D_N along each coordinate j.

Theorem 5.1 (Guaranteed sampled bound)**.** *Let* $\Phi : D \to \mathbb{R}$ *be continuously differentiable, and let*

$$L_j(D) := \sup_{\mathbf{q}\in D} |\partial_j \Phi(\mathbf{q})|, \quad j \in \{1, \ldots, d\}. \tag{5.13}$$

Then

$$\sup_{\mathbf{q}\in D} \Phi(\mathbf{q}) \leq \max_{\mathbf{q}\in D_N} \Phi(\mathbf{q}) + \frac{1}{N} \sum_{j=1}^{d} L_j(D)\, \delta k_j. \tag{5.14}$$

Proof. Let $\mathbf{q}^* \in D$ attain the supremum and $\mathbf{q} \in D_N$ be such that $|q_j^* - q_j| \leq \delta k_j / N$ for all j (such a $\mathbf{q}$ exists by construction of D_N). The mean value theorem, applied along the segment from $\mathbf{q}$ to $\mathbf{q}^*$ (which lies in D by convexity), gives

$$|\Phi(\mathbf{q}^*) - \Phi(\mathbf{q})| \leq \sum_{j=1}^{d} \sup_{\mathbf{q}''\in D} |\partial_j \Phi(\mathbf{q}'')|\, |q_j^* - q_j| \leq \frac{1}{N} \sum_{j=1}^{d} L_j(D)\, \delta k_j. \tag{5.15}$$

Thus $\Phi(\mathbf{q}^*) \leq \Phi(\mathbf{q}) + \frac{1}{N} \sum_j L_j(D)\, \delta k_j \leq \max_{\mathbf{q}'\in D_N} \Phi(\mathbf{q}') + \frac{1}{N} \sum_j L_j(D)\, \delta k_j$, which is (5.14). $\square$

Refined sampling versus subdivision. The bound (5.14) contains two contributions: the sampled maximum, exact at the grid points, and a correction of order $\mathcal{O}(\|\boldsymbol{\delta}\mathbf{k}\|/N)$. As $N \to \infty$ at fixed box, the correction vanishes and the sampled bound converges to the true supremum. A natural alternative to refining the sampling is to bisect D into sub-boxes. Doubling N on D and halving each δk_j reduce the correction by the same factor of 2; the two strategies are essentially equivalent in terms of accuracy. They differ in the surrounding cost:

6 Effective Hamiltonian and Pauli decomposition

When two eigenvalues coincide at a point $\mathbf{k}_c$, the standard Rayleigh–Schrödinger expansion becomes singular: the denominator $\varepsilon_{n,0}-\varepsilon_{m,0}$ vanishes for the crossing pair. As a consequence, the eigenvalues $\varepsilon_n(\mathbf{k})$ generically lose differentiability at $\mathbf{k}_c$, even though $H(\mathbf{k})$ remains smooth (in fact analytic in the usual Bloch setting).

The standard remedy is to project onto the degenerate subspace and to diagonalise the resulting low-dimensional Hamiltonian explicitly. Let $\varepsilon_i(\mathbf{k}_c) = \varepsilon_j(\mathbf{k}_c)$ be a doubly degenerate eigenvalue with associated orthonormal eigenvectors $\psi_i(\mathbf{k}_c), \psi_j(\mathbf{k}_c)$, and let

$$P(\mathbf{k}_c) = \frac{1}{2\pi i} \oint_\Gamma \frac{\mathrm{i}z}{H(\mathbf{k}_c) - z}, \tag{6.1}$$

where Γ is a small positively oriented contour around the degeneracy. Because of the analytic functional calculus, $P(\mathbf{k}_c)$ is the orthogonal projector onto the two-dimensional crossing subspace $\mathcal{E}_c = \mathrm{span}\{\psi_i, \psi_j\}$.

The *effective Hamiltonian* near $\mathbf{k}_c$ is the 2×2 Hermitian matrix

$$H_{\text{eff}}(\mathbf{k}) = P(\mathbf{k}_c)\, H(\mathbf{k})\, P(\mathbf{k}_c), \tag{6.2}$$

read in the fixed basis (ψ_i, ψ_j). Its eigenvalues coincide with the two physical crossing eigenvalues of $H(\mathbf{k})$ to leading order in $\mathbf{k} - \mathbf{k}_c$. Throughout we work in a small box $D = \mathbf{k}_c + [-\delta k, \delta k]^d$ and we assume an *exterior gap* $\gamma > 0$: all other eigenvalues of $H(\mathbf{k})$ stay at distance at least γ from the crossing pair for every $\mathbf{k} \in D$.

6.1 Pauli decomposition

Any 2×2 Hermitian matrix admits a unique decomposition in the basis $(I, \sigma_1, \sigma_2, \sigma_3)$, where the Pauli matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & \mathrm{i} \\ -\mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{6.3}$$

We write

$$H_{\text{eff}}(\mathbf{k}) = a(\mathbf{k})\, I + \mathbf{b}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \qquad a(\mathbf{k}) \in \mathbb{R}, \quad \mathbf{b}(\mathbf{k}) \in \mathbb{R}^3. \tag{6.4}$$

A direct computation gives the eigenvalues

$$\lambda_\pm(\mathbf{k}) = a(\mathbf{k}) \pm \|\mathbf{b}(\mathbf{k})\|_2. \tag{6.5}$$

The two eigenvalues coincide if and only if $\mathbf{b}(\mathbf{k}) = \mathbf{0}$; this is a system of three real equations in the d-imensional parameter $\mathbf{k}$. For a generic Hamiltonian, crossings therefore occur on a set of codimension 3: they are absent in $d \leq 2$ and form isolated points in $d = 3$ (Weyl points).

In real materials, symmetries can constrain some components of $\mathbf{b}$ to vanish identically along high-symmetry submanifolds of the Brillouin zone, lowering the effective codimension of the crossing condition. Crossings of codimension 2 (nodal lines) or codimension 1 (nodal surfaces) then become possible and persist under any symmetry-preserving perturbation of H.

6.2 Linear expansion of the Pauli vector

A first-order Taylor expansion of $\mathbf{b}$ around the crossing point $\mathbf{k}_c$, where $\mathbf{b}(\mathbf{k}_c) = \mathbf{0}$, yields

$$\mathbf{b}(\mathbf{k}) = \sum_{i=1}^{d} \frac{\partial \mathbf{b}(\mathbf{k}_c)}{\partial k_i}\, \delta k_i + \mathcal{O}\big(\|\delta \mathbf{k}\|^2\big), \qquad \delta \mathbf{k} := \mathbf{k} - \mathbf{k}_c. \tag{6.6}$$

It is convenient to introduce the $3 \times d$ Jacobian matrix

$$M_{i,j} := \frac{\partial b_i}{\partial k_j}(\mathbf{k}_c), \qquad i = 1, 2, 3, \quad j = 1, \ldots, d, \tag{6.7}$$

so that, to leading order,

$$\mathbf{b}(\mathbf{k}) = M\, \delta \mathbf{k} + \mathcal{O}\big(\|\delta \mathbf{k}\|^2\big). \tag{6.8}$$

Combined with (6.5), the local description of the two crossing bands is

$$\lambda_\pm(\mathbf{k}) - a(\mathbf{k}_c) \approx \pm \Big\| \sum_{i=1}^{d} \frac{\partial \mathbf{b}(\mathbf{k}_c)}{\partial k_i}\, \delta k_i \Big\|_2 \tag{6.9}$$